

1 Linear Systems

1.1 Vector Norms Vector norm $\|\cdot\|: \mathbb{R}^n \rightarrow \mathbb{R}$. Properties:

- PD: $\|x\| \geq 0, \|x\| = 0 \iff x = 0$
- Homogeneity: $\|\alpha x\| = |\alpha| \cdot \|x\|$
- Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p\right)^{1/p},$$
$$\|x\|_\infty = \max |x_i|$$

1.2 Matrix Norms Matrix norm $\|\cdot\|: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$. Properties:

- PD: $\|A\| \geq 0, \|A\| = 0 \iff A = 0$
- Homogeneity: $\|\alpha A\| = |\alpha| \cdot \|A\|$
- Triangle inequality: $\|A + B\| \leq \|A\| + \|B\|$
- Submultiplicativity: $\|AB\| \leq \|A\| \|B\|$

$$\|A\|_1 = \max_j \sum_{i=1}^n |a_{ij}|,$$
$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)},$$
$$\|A\|_F = \sqrt{\sum_{i,j=1}^n |a_{ij}|^2},$$
$$\|A\|_\infty = \max_i \sum_{j=1}^n |a_{ij}|$$

1.3 Condition Number / Sensitivity:

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

1.4 Gaussian Elimination ($\sim \frac{2}{3}n^3$):

```
for k = 1, ..., n - 1 // pivot el.
  for i = k + 1, ..., n // rows
    xmult = aik / akk
    aik = 0
    for j = k + 1, ..., n // columns
      aij = aij - xmult akj
    end
    bi = bi - xmult bk
  end
end
```

Back Substitution ($\sim n^2$):

```
xn = bn / unn
for i = n - 1, ..., 1
  s = bi
  for j = i + 1, ..., n
    s = s - uij xj
  end
  xi = s / uii
end // solving Ux=b
```

1.5 LU Decomposition Decomposition of $A \in \mathbb{R}^{n \times n}$ into a lower Δ and an upper Δ matrix:

$$A = LU$$

start with $L = I, U = A$. **Algorithm:**

```
for k = 1, ..., n - 1
  for i = k + 1, ..., n
    lik = aik / akk
    for j = k + 1, ..., n
      aij = aij - lik akj
    end
  end
end // same as G.E., but store multipliers in L
```

Solving the system $Ax = b$:

$$Ax = LU \quad (\sim \tfrac{2}{3}n^3)$$
$$Ly = b \quad (\sim n^2)$$
$$Ux = y \quad (\sim n^2)$$

1.6 Pivoting

$$PA = LU$$

Partial pivoting: replace the j -th row with the row having the largest $|a_{ij}|, i = j, \dots, n$.

Permutation matrix: $P^T = P^{-1}$.

$$P = I_{\text{column } i \leftrightarrow j}, \quad PA = A_{\text{row } i \leftrightarrow j}$$

Swap the 2nd and 3rd rows of matrix A :

$$PA, \quad P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Solving the system $Ax = b$:

$$A = P^T LU \quad (\sim \tfrac{2}{3}n^3)$$
$$Ly = Pb \quad (\sim n^2)$$
$$Ux = y \quad (\sim n^2)$$

1.7 Cholesky ($\sim \frac{1}{3}n^3$). LU decomposition for PD matrices:

$$A = LL^T$$

Algorithm:

```
for k = 1, ..., n // go along diagonal
  lkk = sqrt(akk - sum_{i=1}^{k-1} lki}^2)
  for j = k + 1, ..., n // for each el. j below diag.
    for i = 1, ..., k - 1
      ajk = ajk - lji lki
    end
    ljk = ajk / lkk
  end
end
```

Exists when matrix A is:

- Symmetric: $A = A^T$
- PD: $x^T Ax > 0 \quad \forall x \neq 0$

1.8 Spectral Radius.

$$\rho(R) = \max_i |\lambda_i(R)|.$$

$x^{(k+1)} = Rx^{(k)} + c$ converges for every $x^{(0)}$ $\iff \rho(R) < 1$.

Strict row diagonal dominance: $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for every i .

1.9 Jacobi Iteration ($O(mn)$).

$$x_i^{(k)} = \frac{1}{a_{ii}} \left(b_i - \sum_{\substack{j=1 \\ j \neq i}}^n a_{ij} x_j^{(k-1)} \right), \quad i = 1, \dots, n.$$

$$x_n^{(k)} = \frac{b_n - a_{n,1} x_1^{(k-1)} - \dots - a_{n,n-1} x_{n-1}^{(k-1)}}{a_{nn}}$$

1.10 Gauss-Seidel Component form:

$$x_i^{+1} = \frac{1}{a_{ii}} \left(b_i - \sum_{j < i} a_{ij} x_j^{+1} - \sum_{j > i} a_{ij} x_j^k \right)$$

1.11 SOR Relaxed GS iteration with $\omega \in \mathbb{R}$:

$$x_i^{(k)} = (1 - \omega) x_i^{(k-1)} + \omega x_i^{\text{GS}}.$$

$$x_i^{(k)} = (1 - \omega) x_i^{(k-1)} + \frac{\omega}{a_{ii}} \left(b_i - \sum_{j < i} a_{ij} x_j^{(k)} - \sum_{j > i} a_{ij} x_j^{(k-1)} \right).$$

If A is PD and $0 < \omega < 2$, SOR converges.

1.12 Vectors are conjugate, if $x^T Ay = 0$ ($\langle x, y \rangle_A = x^T Ay$).

1.13 Overdetermined Systems more equations than unknowns:

$$A \in \mathbb{R}^{n \times m}, \quad b \in \mathbb{R}^n, \quad n \gg m$$

1.14 Least Squares Method

$$\hat{x} = \underset{x}{\operatorname{argmin}} \|Ax - b\|$$

The least-squares solution of $Ax = b$ is x , which solves the system $A^T Ax = A^T b$. Using Cholesky:

$$B = A^T A, \quad c = A^T b \quad (\sim nm^2)$$
$$B = VV^T \quad (\sim \tfrac{1}{3}m^3)$$
$$Vy = c \quad (\sim m^2)$$
$$V^T \hat{x} = y \quad (\sim m^2)$$

1.15 QR Decomposition For $A \in \mathbb{R}^{n \times m}, n \geq m$

$$A = QR$$

$$Q^T = Q^{-1}, \quad R \text{ upper } \Delta$$

Solve:

$$Rx = Q^T b$$

1.16 Gram-Schmidt Orthogonalization of the set $\{a_1, \dots, a_n\} \Rightarrow \{q_1, \dots, q_n\}$, such that $L(q_i) = L(a_i)$

$$q_1 = \frac{a_1}{\|a_1\|}, \quad q_k = \frac{a_k - \sum_{j=1}^{k-1} \langle a_k, q_j \rangle q_j}{\|q_k\|}$$

1.17 Householder Reflection

$$H_w = I - 2 \frac{ww^T}{w^T w}, \quad H_w^T = H_w, \quad H_w^2 = I$$

$$w = x + \operatorname{sign}(x_1) \|x\|_2 \cdot e_1$$

$$R = H_k \dots H_1 A, \quad Q = H_1 \dots H_k$$

1.18 Singular Value Decomposition (SVD). For $A \in \mathbb{R}^{n \times m}$, there exist orthogonal $U \in \mathbb{R}^{n \times n}$, $V \in \mathbb{R}^{m \times m}$ and diagonal $\Sigma \in \mathbb{R}^{n \times m}$:

$$UU^T = I_n, \quad V^T V = I_m, \quad \Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_{\min(n,m)})$$

$$\sigma_1 \geq \dots \geq \sigma_{\min(n,m)} \geq 0, \quad A = U \Sigma V^T.$$

Right $v_i \in \mathbb{R}^m$ are input directions, and left $u_i \in \mathbb{R}^n$ are output directions: $Av_i = \sigma_i u_i$.

2 Finding Eigenvalues

2.1 Power Method Finds the dominant eigenpair by repeated multiplication:

$$x^{(k+1)} = \frac{Ax^{(k)}}{\|Ax^{(k)}\|}, Ax_0 = \lambda x_0$$

The convergence rate is $\approx \left| \frac{\lambda_2}{\lambda_1} \right|^k$.

2.2 Inverse Iteration For a shift $\sigma \approx \lambda$, use the power method on $(A - \sigma I)^{-1}$; obtain the eigenvalue closest to σ :

$$(A - \sigma I)y^{(k+1)} = x^{(k)}, \quad x^{(k+1)} = \frac{y^{(k+1)}}{\|y^{(k+1)}\|}.$$

2.3 Orthogonal Iteration Generalization of the power method for the first p eigenpairs. For orthonormal $Z_0 \in \mathbb{R}^{n \times p}$, repeat:

$$Y_{k+1} = AZ_k, \quad Y_{k+1} = Z_{k+1} R_{k+1} \quad (\text{QR}).$$

Columns of Z_k converge to the dominant eigenspace. **2.4 QR Iteration** To compute all eigenvalues, repeat:

$$A_0 = A, \quad A_k = Q_k R_k, \quad A_{k+1} = R_k Q_k.$$

The diagonal of A_k converges to the eigenvalues.

3 Solving Nonlinear Equations

3.1 Difference of Function Values When $x_a, x_b \rightarrow x^*$, we have

$$\frac{f(x_b) - f(x_a)}{x_b - x_a} \rightarrow f'(x^*), \quad |f(x_b) - f(x_a)| \approx |f'(x^*)| |x_b - x_a|$$

3.2 Bisection Finding a zero of function $f(x)$ on interval $[a, b]$, where $f(a) \cdot f(b) < 0$.

```
for k = 1, 2, ...
  xm = a + (b - a)/2
  if sign(f(xm)) = sign(f(a))
    a = xm
  else
    b = xm
  end
  if |b - a| < tol, stop
end
```

3.3 Tangent Method / Newton's Method Use the tangent at point x_k to find a zero of $f(x)$:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Point x_{k+1} is the intersection of the tangent at x_k with the x-axis. The order of convergence is $r = 2$, if $f'(x^*) \neq 0$.

3.4 Secant Method Approximate the derivative using the two previous points, so we need two initial approximations: x_0, x_1 :

$$f'(x_k) \approx \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}}$$

$$x_{k+1} = x_k - f(x_k) \cdot \frac{x_k - x_{k-1}}{f(x_k) - f(x_{k-1})}$$

3.5 Newton's Method Generalization of the tangent method for nonlinear systems $F(x) = 0$.

$$J(x_k) \Delta x = -F(x_k)$$

Solve the linear system for Δx by any method, then update:

$$x_{k+1} = x_k + \Delta x$$

$$J(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

3.6 Broyden's Method Approximate the Jacobian matrix by B_r :

$$B_{r+1} = B_r + \frac{F(x^{(r+1)})(\Delta x^{(r)})^T}{\|\Delta x^{(r)}\|_2^2}.$$

3.7 Variational Methods Zeros of $f(x)$ are global minima of the function

$$g(x) = \|f(x)\|^2 = \sum_{i=1}^n (f_i(x))^2.$$

Find the minimum iteratively:

$$x^{(r+1)} = x^{(r)} + \lambda_r v_r, \qquad g(x^{(r+1)}) < g(x^{(r)}).$$

3.8 Choosing the Descent Direction

- general: $v_r \not\perp \nabla g(x)$,
- steepest descent: $v_r = -\nabla g(x)$,
- coordinate descent: $v_r = e_1, e_2, \dots, e_n$ in sequence.

3.9 Gauss–Newton Iteration For an overdetermined nonlinear system, use the pseudoinverse of the Jacobian matrix:

$$x^{(r+1)} = x^{(r)} - J_f(x^{(r)})^+ f(x^{(r)}).$$

4 Polynomial Interpolation

4.1 Problem. Find a polynomial $p(x)$ of degree at most $n-1$ that satisfies:

$$p(x_i) = y_i, \quad \text{for } i = 1, 2, \dots, n$$

This means that the polynomial interpolates the given points (x_i, y_i) .

4.2 Vandermonde System This problem can be written as a linear system:

$$\begin{bmatrix} 1 & x_1 & \cdots & x_1^{n-1} \\ 1 & x_2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & \cdots & x_n^{n-1} \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

4.3 Orthogonal Polynomials The Vandermonde system can be ill-conditioned, so it is better to use an orthogonal basis $g_0, \dots, g_n$:

$$\langle f, g \rangle = \sum_{i=1}^m f(x_i)g(x_i), \qquad \langle g_j, g_k \rangle = 0 \ (j \neq k).$$

$$p(x) = \sum_{i=0}^n a_i g_i(x), \qquad a_i = \frac{\langle f, g_i \rangle}{\langle g_i, g_i \rangle}.$$

4.4 Lagrange Form

$$\ell_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \qquad i = 0, \dots, n.$$

$$p(x) = \sum_{i=0}^n \ell_i(x) y_i.$$

4.5 Newton Form

$$p_n(x) = c_0 + c_1(x - x_0) + \cdots + c_n \prod_{j=0}^{n-1} (x - x_j).$$

The coefficients are divided differences:

$$c_i = f[x_0, \dots, x_i], \qquad i = 0, \dots, n.$$

x	$f[\cdot]$	$f[\cdot, \cdot]$	$f[\cdot, \cdot, \cdot]$
x_0	$f[x_0]$		
x_1	$f[x_1]$	$f[x_0, x_1]$	
x_2	$f[x_2]$	$f[x_1, x_2]$	$f[x_0, x_1, x_2]$
x_3	$f[x_3]$	$f[x_2, x_3]$	$f[x_1, x_2, x_3]$

Newton coefficients are the top diagonal:

$$c_0 = f[x_0], \quad c_1 = f[x_0, x_1], \quad c_2 = f[x_0, x_1, x_2], \dots$$

$$f[x_i] = f(x_i) = y_i, \qquad f[x_i, x_j] = \frac{f(x_j) - f(x_i)}{x_j - x_i}.$$

$$f[x_0, \dots, x_n] = \frac{f[x_1, \dots, x_n] - f[x_0, \dots, x_{n-1}]}{x_n - x_0}.$$

5 Numerical Integration

5.1 Newton–Cotes Rules A special form of least squares, where nodes x_i are chosen equidistantly on interval $[a, b]$:

$$x_i = a + ih, \quad h = \frac{b-a}{mn}, \quad i = 0, \dots, n$$

Trapezoidal Rule ($n = 1, \leq 1$):

$$= \frac{h}{2}(f(a) + f(b)) - \frac{h^3 f''(\xi)}{12}$$

Simpson's Rule ($n = 2, \leq 3!$):

$$= \frac{h}{3}(f(x_0) + 4f(x_1) + f(x_2)) - \frac{h^5 f''(\xi)}{90}$$

Composite Trapezoidal ($m > 1$):

$$= \frac{h}{2}(f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) +$$

$$f(x_n)) - \frac{(b-a)h^2 f''(\xi)}{12}$$

Composite Simpson ($m > 1$):

$$= \frac{h}{3}(f(x_0) + 4f(x_1) + 2f(x_2) + \cdots +$$

$$2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)) - \frac{(b-a)h^4 f''(\xi)}{280}$$

5.2 Romberg Method $h = (b-a)/n$ is the subinterval width of the composite trapezoidal rule $T(h)$. The error has an expansion in even powers:

$$T(h) = I + c_1 h^2 + c_2 h^4 + \cdots$$

In each new row, double the number of subintervals, so $h_\ell = h_0/2^\ell$. Start with

$$R_{\ell,0} = T(h_\ell)$$

and compute the triangular table:

$$R_{\ell,m} = \frac{4^m R_{\ell,m-1} - R_{\ell-1,m-1}}{4^m - 1}, \qquad 1 \leq m \leq \ell.$$

h	$R_{\ell,0} = T(h_\ell)$	$R_{\ell,1}$	$R_{\ell,2}$
h_0	$R_{0,0}$		
$h_0/2$	$R_{1,0}$	$R_{1,1}$	
$h_0/4$	$R_{2,0}$	$R_{2,1}$	$R_{2,2}$
$h_0/8$	$R_{3,0}$	$R_{3,1}$	$R_{3,2}$

Best approximations are on the top diagonal:

$$R_{0,0}, \quad R_{1,1}, \quad R_{2,2}, \dots$$

Each new column eliminates the next error term: $h^2, h^4, \dots$

6 Differential Equations (DE)

6.1 Definition. DE: equation of the form:

$$F(x, y, y', y'', \dots, y^{(n)}) = 0, \quad y = y(x)$$

The general solution is a family of functions with n parameters. A particular solution is obtained when boundary conditions are given.
Order of a DE: the highest derivative that appears.
6.2 Introducing a New Variable. Convert a higher-order ODE to a first-order system by naming each derivative:

$$u_1 = y, \quad u_2 = y', \quad \dots, \quad u_n = y^{(n-1)}.$$

Then

$$u'_1 = u_2, \quad u'_2 = u_3, \quad \dots, \quad u'_n = F(x, u_1, \dots, u_n).$$

Example:

$$y'' = F(x, y, y') \Rightarrow \begin{cases} u'_1 = u_2, \\ u'_2 = F(x, u_1, u_2). \end{cases}$$

For $y'' = y' + 2y$:

$$u_1 = y, \quad u_2 = y', \quad u'_1 = u_2, \quad u'_2 = u_2 + 2u_1.$$

6.3 Local and Global Error. **Local error:** the difference between approximation y_i , obtained from approximation y_{i-1} , and $z(x_i)$, the result if the same step were taken from exact $y(x_{i-1})$.

$$l_i = y_i - z(x_i)$$

Global error: the difference between approximation y_i and exact $y(x_i)$ at x_i :

$$g_i = y_i - y(x_i) \leq \sum |l_j|$$

6.4 Order of a Method $p \in \mathbb{N}$, for which:

$$l_i = Ch^{p+1} + O(h^{p+2})$$

The order of a method determines the accuracy of the approximation (higher $p \rightarrow$ better method).

6.5 Euler's Method Order 1: linear approx. using the tangent at x_i .

Explicit:

$$y_{i+1} = y_i + h_i \cdot f(x_i, y_i)$$

Implicit:

$$y_{i+1} = y_i + h_i \cdot f(x_{i+1}, y_{i+1})$$

6.6 Runge–Kutta 4 (RK4) For the IVP $y' = f(x, y)$, $y(x_0) = y_0$ and step h :

$$k_1 = f(x_n, y_n)$$

$$k_2 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2\right)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Idea: estimate the slope at start, two midpoints, and end; then use the weighted average $1 : 2 : 2 : 1$.

$$x_{n+1} = x_n + h$$

Order 4: local error $O(h^5)$, global error $O(h^4)$. For systems, use the same formulas with vector y_n and slopes k_i .